

Internship Progress Report

Chronic Disease Burden & Social Determinants Dashboard

Date: April 8, 2026
Supervisor: Richard Rothenberg
Organization: Global AI
Period: Week 4, Part-Time Internship

Work Completed

This week I moved into the segmentation phase of the project. The goal was to group U.S. counties into meaningful risk tiers based on their combined chronic disease burden and social vulnerability profile, then visualise the results in a way that is actually interpretable rather than just technically correct. It took most of the week to get there, but the output is four figures and a saved CSV of clustered counties.

Feature Selection for Clustering

Before running any clustering algorithm I needed to decide which features to include. Rather than selecting them by hand, I used the correlation results from Week 2 as a data-driven filter: any factor with an average absolute Pearson correlation of at least 0.40 across both diabetes and hypertension was included. This gave a final feature set of 16 variables covering both disease outcomes, material deprivation measures (food insecurity, housing insecurity, lack of transportation), behavioural SDOH factors (physical inactivity, insufficient sleep, loneliness, obesity, current smoking), and SVI components (overall vulnerability, socioeconomic vulnerability, household characteristics, racial/ethnic minority status). Binge drinking was retained despite its negative correlation because it had sufficient average magnitude to pass the threshold and its directional behaviour across tiers was worth examining.

All features were standardised using `StandardScaler` before fitting so that no single variable dominated the distance calculations due to differences in scale.

Choosing the Number of Clusters

I ran k-means for $K = 2$ through $K = 8$ and evaluated each solution using both inertia and silhouette score (Figure 1). The elbow plot shows a clear bend at $K = 4$, where the rate of inertia reduction flattens noticeably. The silhouette score is consistent with this: it peaks at $K = 2$ but drops sharply by $K = 3$ and then levels off from $K = 4$ onward, meaning that going beyond four clusters does not meaningfully improve separation. Four clusters also maps cleanly onto the Low, Moderate, High, and Critical tier labels that make sense for a public health dashboard, so $K = 4$ was the chosen solution.

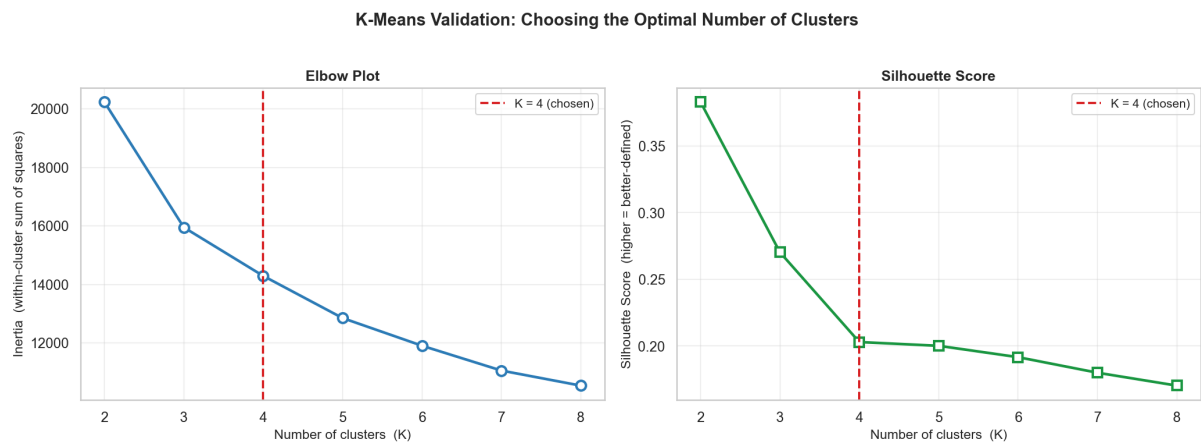


Figure 1: Elbow plot (left) and silhouette score (right) for $K = 2$ to $K = 8$. The red dashed line marks the chosen solution at $K = 4$.

Cluster Profiles

After fitting the final four-cluster model, clusters were ranked by composite risk (mean of all standardised features) and labelled Low, Moderate, High, and Critical in ascending order. The profile heatmap in Figure 2 shows the mean value of each feature per tier. The gradient is clean and monotone across almost every column, which gives some confidence that the clusters are capturing a genuine underlying risk dimension rather than just splitting on noise.

The Critical tier averages 15.8% diabetes prevalence, 42.9% hypertension, and an SVI overall score of 0.9, compared to 9.1%, 29.8%, and 0.2 respectively in the Low tier. Food insecurity, housing insecurity, and lack of transportation all follow the same monotone pattern across tiers, which is consistent with the Week 2 finding that material deprivation is the strongest county-level predictor of chronic disease burden.

Binge drinking is the one column that runs in reverse: the Low tier has the highest mean (20.0) and the Critical tier the lowest (14.2). This is the same confounding pattern flagged in Week 2 and it holds within the cluster structure, so it is not an artefact of the correlation analysis.

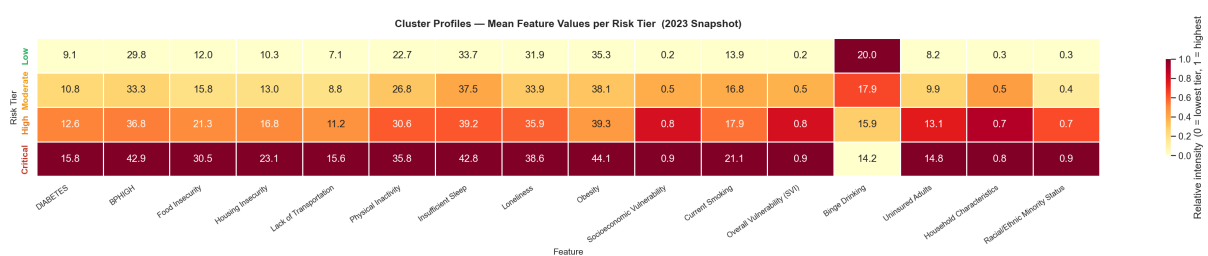


Figure 2: Mean feature values per risk tier for the 2023 snapshot. Colour intensity is normalised column-wise (0 = lowest tier mean, 1 = highest). Tier labels on the y-axis are colour-coded to match the map legend.

Feature Distributions by Tier

Figure 3 shows boxplots of every clustering feature broken down by tier. The separation is generally clean for the disease outcomes and material deprivation measures. Physical inactivity and insufficient sleep also show well-separated distributions. Racial/ethnic minority status and household characteristics have wider boxes with more overlap between tiers, which suggests they contribute less cleanly to the

separation even though they passed the correlation threshold. That is worth keeping in mind when interpreting tier assignments for individual counties.

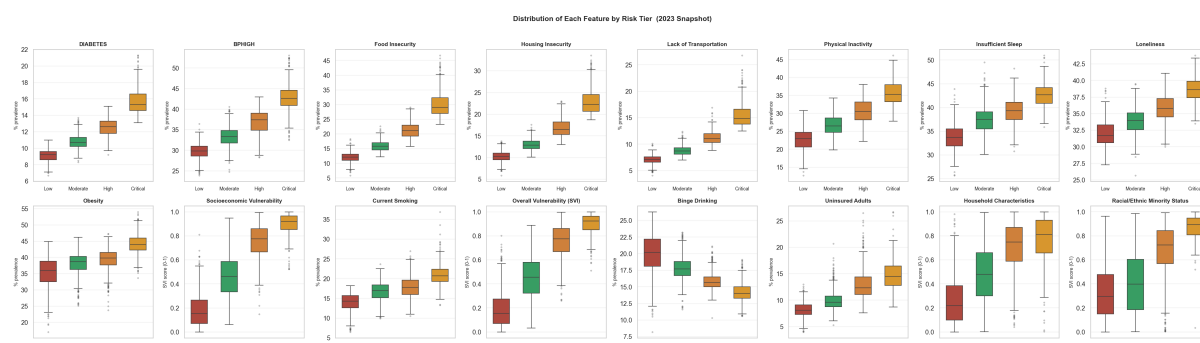


Figure 3: Distribution of each clustering feature by risk tier. Boxes show the interquartile range; dots are outliers.

Risk Tier Map

The final output is an interactive choropleth map (results/fig9_risk_tier_map.html) showing each county colour-coded by its assigned risk tier: green for Low, amber for Moderate, orange for High, and red for Critical. Hovering over any county shows its tier, diabetes and hypertension prevalence, and overall SVI score. The geographic pattern is consistent with expectations: Critical counties are concentrated in the Deep South, parts of Appalachia, and the southern border region, while Low counties cluster in the upper Midwest, Mountain West, and suburban Northeast.

Difficulties Encountered

- Choosing the correlation threshold for feature selection.** The 0.40 cutoff was a judgement call. A lower threshold would have added weakly correlated features that could blur the cluster structure; a higher one would have excluded variables like racial/ethnic minority status that are substantively important even if their county-level correlation is moderate. I settled on 0.40 as a reasonable balance but it is not the only defensible choice.
- Silhouette scores are lower than expected.** The silhouette score at $K = 4$ is around 0.20, which is on the low end for a well-separated clustering solution. This is partly a consequence of using 16 features in a high-dimensional space where Euclidean distance becomes less discriminative, and partly because county-level health data genuinely lies on a continuum rather than forming tight discrete clusters. The tiers are still interpretable and internally consistent, but the low silhouette score is worth being transparent about.
- Assigning tier labels after fitting.** K-means does not return ordered labels, so I had to rank the four clusters by composite risk after fitting and map them onto the Low through Critical labels. The ranking was stable across multiple runs with different random seeds, which gave me confidence it was not sensitive to initialisation.

Planned Work

The analytical work is now largely complete. Next week I plan to write the one-page executive brief summarising the key findings across all four weeks for Richard, covering the main correlation results, what the four risk tiers represent, and what the maps show geographically. I also want to do a final check of the pipeline end to end to make sure all output files are consistently named and saved before wrapping up.